

Graphs and Graphing Algorithms

```
In [ ]: from jupyterquiz import display_quiz  
quiz_path = "questions/ch8/"
```

1. The Graph Abstract Data Type
2. Breadth-First Search
3. Depth-First Search
4. Shortest Path Problems
5. Minimum Spanning Tree

9.1 Introduction

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We will see in this chapter that once we have a good representation for a problem, we can use some standard graph algorithms to solve what otherwise might seem to be a very difficult problem!

A graph is just like a road map. If you have ever used one of the internet map sites, you know that a computer can find the shortest, quickest, or easiest path from one place to another.

9.2. Vocabulary and Definitions

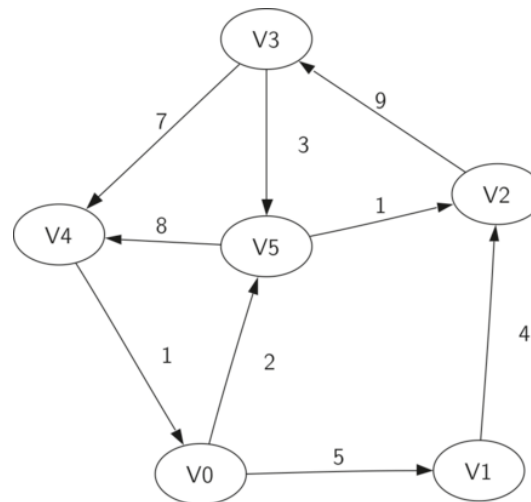
- Vertex: A vertex (also called a **node**) is a fundamental part of a graph. It can have a name, which we will call the **key**. A vertex may also have additional information. We will call this additional information the **value** or the payload.

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- Edge: An edge (also called an **arc**) is another fundamental part of a graph. An edge connects two vertices to show that there is a relationship between them. Edges may be one-way or two-way. If the edges in a graph are all one-way, we say that the graph is a directed graph, or a digraph.

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- Weight: Edges may be weighted to show that there is a cost to go from one vertex to another, which we call **edge cost**. For example, in a graph of roads that connect one city to another, the weight on the edge might represent the distance between the two cities.

A graph can be represented by $G = (V, E)$. For the graph G , V is a set of vertices and E is a set of edges. Each edge is a tuple (v, w) where $v, w \in V$. We can add a third component to the edge tuple to represent a weight. A subgraph s is a set of edges e and vertices v such that $e \subset E$ and $v \subset V$.

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$$V = \{v_0, v_1, v_2, v_3, v_4, v_5\}$$

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$$V = \{v_0, v_1, v_2, v_3, v_4, v_5\}$$

and the set of nine edges:

$$E = \left\{ \begin{array}{l} (v_0, v_1, 5), (v_1, v_2, 4), (v_2, v_3, 9), \\ (v_3, v_4, 7), (v_4, v_0, 1), (v_0, v_5, 2), \\ (v_5, v_4, 8), (v_3, v_5, 3), (v_5, v_2, 1) \end{array} \right\}$$

The example graph helps illustrate two other key graph terms:

- Path: A path in a graph is a sequence of distinct vertices $w_1, w_2, \dots, w_n$ such that $(w_i, w_{i+1}) \in E$ for all $1 \leq i \leq n - 1$. The **unweighted path length** is the number of edges, $n - 1$; the **weighted path length** is the sum of the weights of all edges in the path. For example, the path from v_3 to v_1 is (v_3, v_4, v_0, v_1) . (A more general sequence that allows repeated vertices is called a **walk**.)

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- Cycle: A cycle is a closed sequence $(w_1, w_2, \dots, w_n, w_1)$ where $w_1, \dots, w_n$ are distinct vertices and consecutive vertices are connected by an edge. For example, (v_5, v_2, v_3, v_5) is a cycle. A graph with no cycles is called an **acyclic graph**, and a directed graph with no cycles is called a **directed acyclic graph (DAG)**. We will see that many important problems can be solved efficiently if they can be modeled as a DAG!

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A tree is a connected, acyclic, undirected graph. In other words, there is a path between every pair of vertices, and the graph contains no cycles. We will explore different types of trees in detail in the next chapter!

9.3. The Graph Abstract Data Type

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- `Graph()` : creates an empty graph.
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- `Graph()` : creates an empty graph.
- `setVertex(key)` : inserts a missing vertex. If the key already exists, it is a no-op, so existing edges and traversal state are preserved.
- `addEdge(fromKey, toKey)` : adds a directed, zero-weight edge and creates missing endpoint vertices.
- `addEdge(fromKey, toKey, weight)` : adds or updates a weighted directed edge.

- `vertices.find(key)` or `vertices.count(key)`: tests whether a key exists.
- A range-based loop over `vertices` visits all key/`Vertex` pairs.

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Now that we have looked at the definition for the graph ADT, there are several ways we can implement it. There are two well-known implementations of a graph, the adjacency matrix and the adjacency list.

9.4. An Adjacency Matrix

One of the easiest ways to implement a graph is to use a two-dimensional matrix. Each of the rows and columns represents a vertex in the graph. The value that is stored in the cell at the intersection of row v and column w indicates if there is an edge.

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	V0	V1	V2	V3	V4	V5
V0		5				2
V1			4			
V2				9		
V3					7	3
V4	1					
V5			1		8	

When two vertices are connected by an edge, we say that they are adjacent and the value in each cell represents the weight. The advantage of the adjacency matrix is that it is simple, and for small graphs it is easy to see which nodes are connected to other nodes. However, we can say that this matrix is **sparse**. A matrix is not a very efficient way to store sparse data.

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However, there are few real problems that approach this sort of connectivity. The problems we will look at in this chapter all involve graphs that are sparsely connected!

9.5. An Adjacency List

A sparse graph is stored as an adjacency structure. In our C++ implementation, `Graph::vertices` is a `map<string, Vertex>`, and each `Vertex::neighbors` is a `map<string, int>` from neighbor key to edge weight.

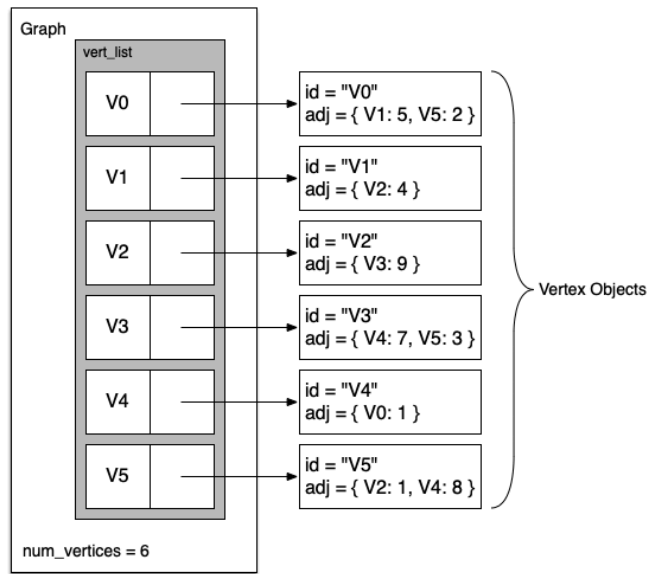
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An adjacency structure uses $O(|V| + |E|)$ storage and lists the neighbors of v in $O(\deg(v))$ time. Our ordered-map implementation additionally makes vertex and edge-key lookup $O(\log |V|)$ and $O(\log \deg(v))$, respectively; those logarithmic factors come from `std::map`, not from the graph ADT itself.



```
In [ ]: display_quiz(quiz_path+"graph.json", max_width=800)
```

Regarding graph implementation methods, which of the following statements is true according to the materials?

- ☐ An adjacency list is more space-efficient for graphs that are sparsely connected.
- ☐ An adjacency matrix is the most space-efficient way to represent a sparse graph.
- ☐ Finding all neighbors of a vertex is faster in an adjacency matrix than in an adjacency list.
- ☐ In an adjacency matrix, if two vertices are not connected, no memory is allocated for that pair.

9.6. Implementation

```

class Vertex {
public:
    string key;
    map<string, int> neighbors;    // neighbor key -> edge weight

    Vertex() {}
    Vertex(string k) {
        key = k;
    }

    int getNeighbor(string other) {
        if (neighbors.count(other)) {
            return neighbors[other];
        }
        return -1;    // no such edge
    }

    void setNeighbor(string other, int weight = 0) {
        neighbors[other] = weight;
    }
};

```

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    void setNeighbor(string other, int weight = 0) {
        neighbors[other] = weight;
    }
};

```

Each `Vertex` uses `map<string, int> neighbors` to associate a neighbor key with an edge weight. `setNeighbor(other, weight)` inserts or updates that mapping, and `getNeighbor(other)` retrieves the weight when the edge exists.

```

class Graph {
    public:
        map<string, Vertex> vertices;

        void setVertex(string key) {
            if (vertices.count(key) == 0) vertices.emplace(key,
Vertex(key));
        }
        void addEdge(string fromVert, string toVert, int weight = 0) {
            if (vertices.count(fromVert) == 0) setVertex(fromVert);
            if (vertices.count(toVert) == 0) setVertex(toVert);
            vertices[fromVert].neighbors[toVert] = weight;
        }
        bool contains(string key) {
            return vertices.count(key) > 0;
        }
};

```



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public:
    map<string, Vertex> vertices;

    void setVertex(string key) {
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    }
    void addEdge(string fromVert, string toVert, int weight = 0) {
        if (vertices.count(fromVert) == 0) setVertex(fromVert);
        if (vertices.count(toVert) == 0) setVertex(toVert);
        vertices[fromVert].neighbors[toVert] = weight;
    }
    bool contains(string key) {
        return vertices.count(key) > 0;
    }
};

```

The `Graph` class also uses `map<string, Vertex> vertices` to map vertex keys to objects. C++ range-based loops iterate over its key/value pairs; `vertices.count(key)` tests membership.

Now let us define the graph seen earlier. We create six vertices numbered 0 through 5, then inspect the `vertices` map. Each key names one `Vertex` object.

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```
In [1]: #include <iostream>
#include "pythonds3/cppds/graph.hpp"    // Vertex + Graph (md Listing above)
using namespace std;

int main() {
    Graph g;
    for (int i = 0; i < 6; i++) {
        g.setVertex(to_string(i));
    }
    for (auto& p : g.vertices) {
        cout << "Vertex(" << p.first << ") ";
    }
    cout << endl;
    return 0;
}
```

Vertex(0) Vertex(1) Vertex(2) Vertex(3) Vertex(4) Vertex(5)

Next, we add the edges that connect the vertices together. Finally, a nested loop verifies that each edge in the graph is properly stored:

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In [2]: #include <iostream>
#include "pythonds3/cppds/graph.hpp"
using namespace std;

int main() {
    Graph g;
    g.addEdge("0","1",5); g.addEdge("0","5",2); g.addEdge("1","2",4);
    g.addEdge("2","3",9); g.addEdge("3","4",7); g.addEdge("3","5",3);
    g.addEdge("4","0",1); g.addEdge("5","4",8); g.addEdge("5","2",1);

    for (auto& p : g.vertices)
        for (auto& n : p.second.neighbors)
            cout << "(" << p.first << "," << n.first << "," << n.second << " "
    cout << endl;
    return 0;
}
```

(0,1,5) (0,5,2) (1,2,4) (2,3,9) (3,4,7) (3,5,3) (4,0,1) (5,2,1) (5,4,8)

9.7. The Word Ladder Problem

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FOOL
POOL
POLL
POLE
PALE
SALE
SAGE

There are many variations of the word ladder puzzle. For example you might be given a particular number of steps in which to accomplish the transformation, or you might need to use a particular word. In this section, we are interested in figuring out the smallest number of transformations needed to turn the starting word into the ending word.

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Here is an outline of where we are going:

- Represent the relationships between the words as a graph.
- Use the graph algorithm known as breadth-first search to find an efficient path from the starting word to the ending word.

9.8. Building the Word Ladder Graph

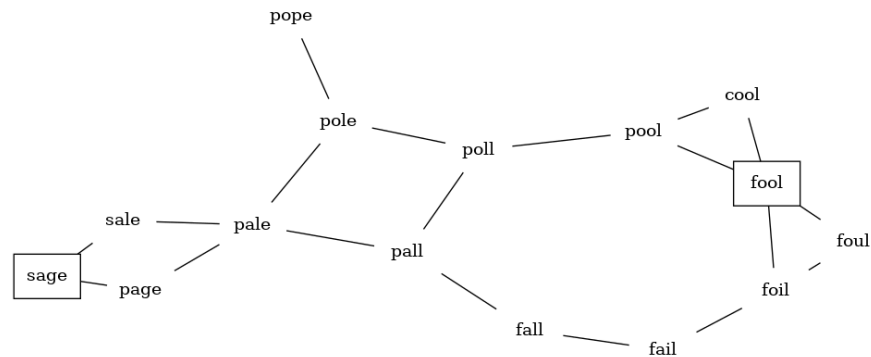
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For a small set of words that approach would work fine. However, let's suppose we have a list of 5,110 words. Roughly speaking, comparing one word to every other word on the list is an $O(n^2)$ algorithm. For 5,110 words, is more than 26 million comparisons!

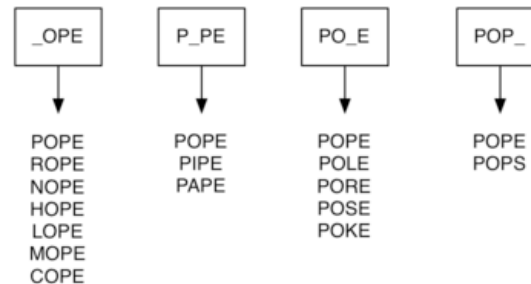
We can do much better by assuming that we have a number of buckets, each labeled with a four-letter word, except that one of the letters on the label has been replaced by an underscore. As we process a list of words, we compare each word with each bucket using the underscore (`_`) as a wildcard.

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Every time we find a matching bucket we put the word in that bucket, so that both `POPE` and `POPS` would both go into the `POP_` bucket. Once we have all the words in the appropriate buckets, we know that all the words in each bucket must be connected!

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The reusable implementation used by the executable examples is split between `pythonds3/cppds/graph.hpp` (data structure) and `pythonds3/cppds/graph_algos.hpp` (algorithms). The longer `graphs/adjacency_graph.cpp` file is retained as a standalone reference listing.

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```
Graph buildGraph(vector<string> words) {
    map<string, set<string>> buckets;
    Graph theGraph;

    // create buckets of words that differ by one letter
    for (string word : words) {
        for (unsigned i = 0; i < word.length(); i++) {
            string bucket = word.substr(0, i) + "_" + word.substr(i + 1);
            buckets[bucket].insert(word);
        }
    }

    // add edges between different words in the same bucket
    for (auto& pair : buckets) {
        for (const string& word1 : pair.second) {
            for (const string& word2 : pair.second) {
                if (word1 != word2) {
                    theGraph.addEdge(word1, word2);
                }
            }
        }
    }
    return theGraph;
}
```


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The 5,110-word adjacency matrix would contain $5,110^2 = 26,112,100$ cells. `buildGraph()` stores 53,286 directed adjacency entries (26,643 undirected word pairs), only about 0.20% of those cells, so the graph is very sparse.

9.9. Implementing Breadth-First Search

With the graph constructed we can now turn our attention to the algorithm we will use to find the shortest solution to the word ladder problem. The graph algorithm we are going to use is called the breadth-first search (BFS), and it is one of the easiest algorithms for searching a graph!

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Given a starting vertex s of a graph, a breadth first search proceeds by exploring edges in the graph to find all the vertices in G for which there is a path from s .

The remarkable thing about a breadth-first search is that it finds all the vertices that are a distance k from s before it finds any vertices that are a distance $k + 1$.

To keep track of its progress, BFS colors each of the vertices white, gray, or black. All the vertices are initialized to white when they are constructed. A white vertex is an undiscovered vertex. When a vertex is initially discovered it is colored gray, and when BFS has completely explored a vertex it is colored black.

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This means that once a vertex is colored black, it has no white vertices adjacent to it. A gray node, on the other hand, may have some white vertices adjacent to it, indicating that there are still additional vertices to explore.

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This means that once a vertex is colored black, it has no white vertices adjacent to it. A gray node, on the other hand, may have some white vertices adjacent to it, indicating that there are still additional vertices to explore.

The `bfs()` shown below uses the adjacency list graph representation we developed earlier. In addition it uses a `Queue`, a crucial point as we will see, to decide which vertex to explore next.

```

void bfs(Graph& g, string startKey) {
    g.vertices[startKey].distance = 0;
    g.vertices[startKey].previous = "";
    queue<string> vertQueue;
    vertQueue.push(startKey);
    while (!vertQueue.empty()) {
        string currentKey = vertQueue.front();
        vertQueue.pop();
        Vertex& current = g.vertices[currentKey];
        for (auto& n : current.neighbors) {
            Vertex& neighbor = g.vertices[n.first];
            if (neighbor.color == "white") {
                neighbor.color = "gray";
                neighbor.distance = current.distance + 1;
                neighbor.previous = currentKey;
                vertQueue.push(n.first);
            }
        }
        current.color = "black";
    }
}

```

```

void bfs(Graph& g, string startKey) {
    g.vertices[startKey].distance = 0;
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    queue<string> vertQueue;
    vertQueue.push(startKey);
    while (!vertQueue.empty()) {
        string currentKey = vertQueue.front();
        vertQueue.pop();
        Vertex& current = g.vertices[currentKey];
        for (auto& n : current.neighbors) {
            Vertex& neighbor = g.vertices[n.first];
            if (neighbor.color == "white") {
                neighbor.color = "gray";
                neighbor.distance = current.distance + 1;
                neighbor.previous = currentKey;
                vertQueue.push(n.first);
            }
        }
        current.color = "black";
    }
}

```

The BFS algorithm uses an extended version of the `Vertex` class that adds three new instance variables: `distance`, `previous`, and `color`. Each of these instance variables also has the appropriate getter and setter methods.

Before each BFS run, every vertex is reset to white, unreachable distance (`INT_MAX`), and no predecessor (empty string). BFS then sets `start.distance = 0`, paints `start` gray **before enqueueing it**, and explores vertices from the front of a FIFO queue.

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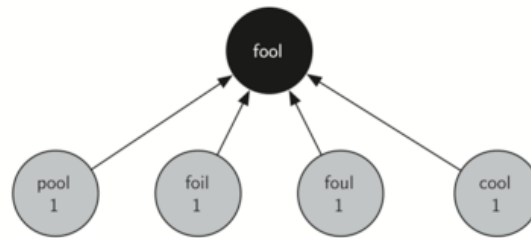
As each node on the adjacency list is examined, its color is checked. If it is white four things happen:

1. The new unexplored vertex `neighbor` is colored gray.
2. The predecessor of `neighbor` is set to the current node `current`.

Before each BFS run, every vertex is reset to white, unreachable distance (`INT_MAX`), and no predecessor (empty string). BFS then sets `start.distance = 0`, paints `start` gray **before enqueueing it**, and explores vertices from the front of a FIFO queue.

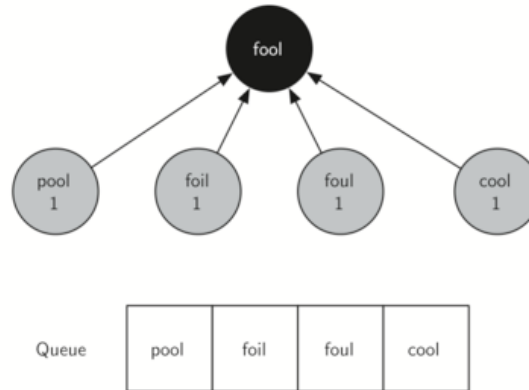
As each node on the adjacency list is examined, its color is checked. If it is white four things happen:

1. The new unexplored vertex `neighbor` is colored gray.
2. The predecessor of `neighbor` is set to the current node `current`.
3. The distance to `neighbor` is set to `current.distance + 1`.
4. `neighbor` is added to the end of the queue. This schedules it after every vertex already waiting at the same or an earlier layer.

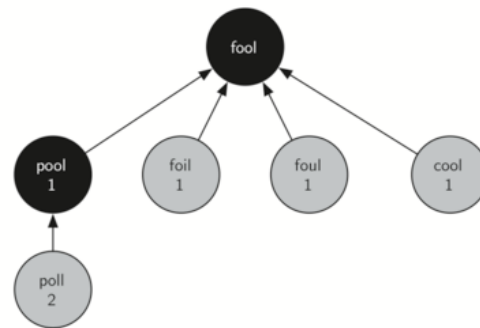


Queue

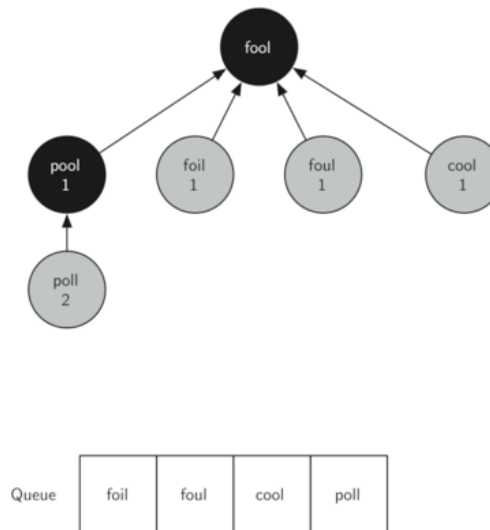
pool	foil	foul	cool
------	------	------	------



Starting from **F00L** we take all nodes that are adjacent to **F00L** and add them to the queue. The adjacent nodes include **P00L**, **FOIL**, **FOUL**, and **COOL**. Each of these nodes are new nodes to expand.



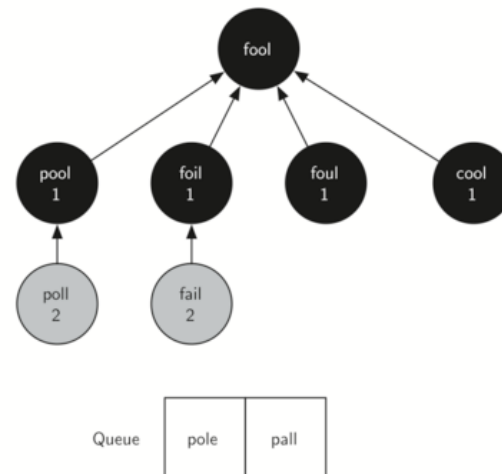
Queue				
	foil	foul	cool	poll

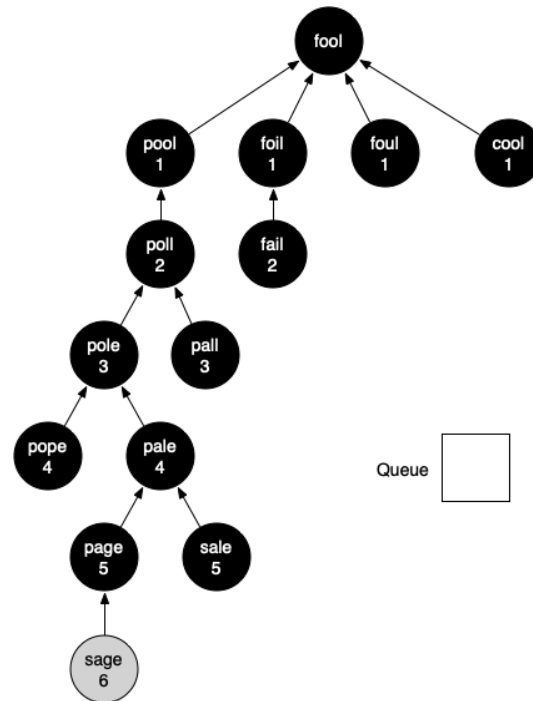


In the next step bfs removes the next node (**PPOOL**) from the front of the queue and repeats the process for all of its adjacent nodes.

The next vertex on the queue is **FOIL**. The only new node that **FOIL** can add to the tree is **FAIL**. As bfs continues to process the queue, neither of the next two nodes adds anything new to the queue or the tree. Below shows the tree and the queue after expanding all the vertices on the second level of the tree:

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We can start at any vertex in the breadth-first search tree and follow the predecessor arrows back to the root to find the shortest word ladder from any word back to `FOOL`.

```
void traverse(Graph& g, string startingKey) {
    string current = startingKey;
    while (current != "") {
        cout << current;
        if (g.vertices[current].previous != "") {
            cout << "->";
        }
        current = g.vertices[current].previous;
    }
    cout << endl;
}
```



```
In [3]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // buildGraph + bfs + traverse
using namespace std;

int main() {
    Graph g = buildGraph({"fool", "cool", "pool", "poll", "pole",
                          "pall", "fall", "fail", "foil", "foul",
                          "pope", "pale", "sale", "sage", "page"});
    g.vertices["fool"].color = "gray";
    bfs(g, "fool");
    traverse(g, "sage");
    return 0;
}
```

sage->page->pale->pall->poll->pool->fool


```
In [3]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // buildGraph + bfs + traverse
using namespace std;

int main() {
    Graph g = buildGraph({"fool", "cool", "pool", "poll", "pole",
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                          "pope", "pale", "sale", "sage", "page"});
    g.vertices["fool"].color = "gray";
    bfs(g, "fool");
    traverse(g, "sage");
    return 0;
}
```

sage->page->pale->pall->poll->pool->fool

```
In [4]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp"
using namespace std;

int main() {
    Graph g = buildGraph({"fool", "cool", "pool", "poll", "pole",
                          "pall", "fall", "fail", "foil", "foul",
                          "pope", "pale", "sale", "sage", "page"});

    bfs(g, "fool");
    for (auto& p : g.vertices) // word(distance from fool)
        cout << p.first << "(" << p.second.distance << ")" ";
    cout << endl;
}
```

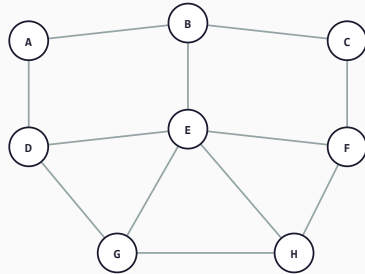
```
return 0;
```

```
}
```

cool(1) fail(2) fall(3) foil(1) fool(0) foul(1) page(5) pale(4) pall(3) pole(3) poll(2) pool(1)
pope(4) sage(6) sale(5)

```
In [ ]: load_anim("bfs")
```

BFS 廣度優先搜尋 —— 層一層往外擴散



› 選擇起點，按「開始」執行 BFS

起點

A ▼

速度



▶ 開始

→ 單步

↺ 重置

Queue : empty

頂點狀態表

v	color	d	prev
A	white	∞	—
B	white	∞	—
C	white	∞	—
D	white	∞	—
E	white	∞	—
F	white	∞	—
G	white	∞	—
H	white	∞	—

9.10. Breadth-First Search Analysis

Let's analyze the run time performance of the breadth-first search algorithm. The first thing to observe is that the `while` loop is executed, at most, one time for each vertex in the graph (up to $|V|$ iterations). You can see that this is true because a vertex must be white before it can be examined and added to the queue.

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This gives us $O(|V|)$ for the while loop. The `for` loop, which is nested inside the `while`, is executed at most once for each edge in the graph (up to $|E|$ iterations). The reason is that every vertex is dequeued at most once and we examine an edge from node u to node v only when node u is dequeued.

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This gives us $O(|E|)$ for the `for` loop. Combining the two loops gives us $O(|V| + |E|)$.

Of course doing the breadth-first search is only part of the task. Following the links from the starting node to the goal node is the other part of the task.

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The worst case for this would be if the graph was a single long chain. In this case traversing through all of the vertices would be $O(|V|)$. The normal case is going to be some fraction of $|V|$.

Finally, at least for this problem, there is the time required to build the initial graph.

9.11. The Knight's Tour Problem

Another classic problem that we can use to illustrate a second common graph algorithm is called the knight's tour. The knight's tour puzzle is played on a chess board with a single chess piece, the knight.

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The object of the puzzle is to find a sequence of moves that allow the knight to visit every square on the board exactly once. One such sequence is called a **tour**.

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The object of the puzzle is to find a sequence of moves that allow the knight to visit every square on the board exactly once. One such sequence is called a **tour**.

The upper bound on the number of possible legal tours for an 8×8 chessboard is known to be 10^{35} ; however, there are even more possible dead ends.

Although researchers have studied many different algorithms to solve the knight's tour problem, a graph search is one of the easiest to understand and program. Once again we will solve the problem using two main steps:

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- Represent the legal moves of a knight on a chessboard as a graph.

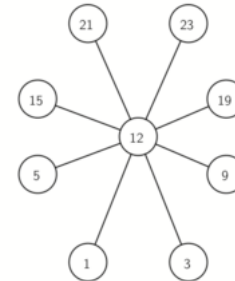
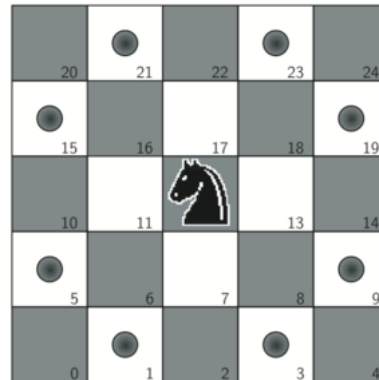
Although researchers have studied many different algorithms to solve the knight's tour problem, a graph search is one of the easiest to understand and program. Once again we will solve the problem using two main steps:

- Represent the legal moves of a knight on a chessboard as a graph.
- Use a graph algorithm to find a path of length $rows \times columns - 1$ where every vertex on the graph is visited exactly once!

9.12. Building the Knight's Tour Graph

To represent the knight's tour problem as a graph we will use the following two ideas: each square on the chessboard can be represented as a node in the graph and each legal move by the knight can be represented as an edge in the graph.

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```

Graph knightGraph(int boardSize) {
    Graph ktGraph;
    for (int row = 0; row < boardSize; row++) {
        for (int col = 0; col < boardSize; col++) {
            int nodeId = row * boardSize + col;
            for (auto& move : genLegalMoves(row, col, boardSize)) {
                int otherNodeId = move.first * boardSize + move.second;
                ktGraph.addEdge(to_string(nodeId), to_string(otherNodeId));
            }
        }
    }
    return ktGraph;
}

```

```

Graph knightGraph(int boardSize) {
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    for (int row = 0; row < boardSize; row++) {
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                ktGraph.addEdge(to_string(nodeId), to_string(otherNodeId));
            }
        }
    }
    return ktGraph;
}

```

The `knightGraph()` function makes one pass over the board. At each square it calls `genLegalMoves()` to produce the legal destination coordinates, then stores one directed adjacency entry for each legal move.


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Graph knightGraph(int boardSize) {
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            for (auto& move : genLegalMoves(row, col, boardSize)) {
                int otherNodeId = move.first * boardSize + move.second;
                ktGraph.addEdge(to_string(nodeId), to_string(otherNodeId));
            }
        }
    }
    return ktGraph;
}

```

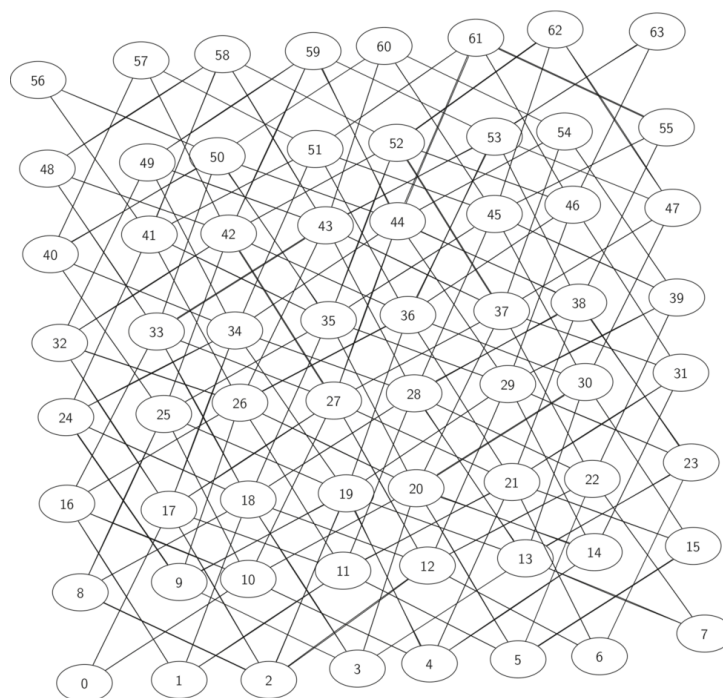
The `knightGraph()` function makes one pass over the board. At each square it calls `genLegalMoves()` to produce the legal destination coordinates, then stores one directed adjacency entry for each legal move.

All legal moves are then converted into edges in the graph. Each location on the board is converted into a linear vertex number.

The `gen_legal_moves()` takes the position of the knight on the board and generates each of the eight possible moves, making sure those moves are still within the board:

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```
vector<pair<int, int>> genLegalMoves(int row, int col, int boardSize) {
    vector<pair<int, int>> newMoves;
    vector<pair<int, int>> moveOffsets = {
        {-1, -2}, {-1, 2}, {-2, -1}, {-2, 1},
        {1, -2}, {1, 2}, {2, -1}, {2, 1}};
    for (auto& off : moveOffsets) {
        int newRow = row + off.first;
        int newCol = col + off.second;
        if (0 <= newRow && newRow < boardSize && 0 <= newCol && newCol <
boardSize) {
            newMoves.push_back({newRow, newCol});
        }
    }
    return newMoves;
}
```



The figure shows the complete graph of possible moves on an 8×8 board. There are exactly 336 edges in the graph. Notice that the vertices corresponding to the edges of the board have fewer connections (legal moves) than the vertices in the middle of the board.

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The 8×8 adjacency matrix has 4,096 cells. The representation stores 336 **directed adjacency arcs** (each legal undirected move appears in both directions), equivalent to 168 undirected edges, so only 8.2% of the matrix cells would be occupied.

9.13. Implementing Knight's Tour

The search algorithm we will use to solve the knight's tour problem is called depth-first search (DFS). Whereas the breadth-first search algorithm builds a **search tree** one level at a time, a depth-first search creates a search tree by exploring one branch of the tree as deeply as possible.

The search algorithm we will use to solve the knight's tour problem is called depth-first search (DFS). Whereas the breadth-first search algorithm builds a **search tree** one level at a time, a depth-first search creates a search tree by exploring one branch of the tree as deeply as possible.

We will look at two algorithms that implement DFS. The first algorithm we will look at specifically solves the knight's tour problem by explicitly **forbidding a node to be visited more than once**.

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We will look at two algorithms that implement DFS. The first algorithm we will look at specifically solves the knight's tour problem by explicitly **forbidding a node to be visited more than once**.

The second implementation is more general, but allows nodes to be visited more than once as the tree is constructed. The second version is used in subsequent sections to develop additional graph algorithms.

The depth-first exploration of the graph is exactly what we need in order to find a path through 64 vertices (one for each square on the chessboard) and 63 edges.

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We will see that when the depth-first search algorithm finds a dead end (a place in the graph where there are no more moves possible) it backs up the tree to the next deepest vertex that allows it to make a legal move.

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We will see that when the depth-first search algorithm finds a dead end (a place in the graph where there are no more moves possible) it backs up the tree to the next deepest vertex that allows it to make a legal move.

The `knightTour()` function takes four main arguments: `n`, the current depth; `path`, a `vector<string>` of visited vertex keys; `uKey`, the vertex being explored; and `limit`, the final path depth.

```

bool knightTour(int n, vector<string>& path, string uKey, int limit, Graph&
g) {
    g.vertices[uKey].color = "gray";
    path.push_back(uKey);
    if (n < limit) {
        bool done = false;
        for (auto& nb : g.vertices[uKey].neighbors) {
            if (done) break;
            if (g.vertices[nb.first].color == "white") {
                done = knightTour(n + 1, path, nb.first, limit, g);
            }
        }
        if (!done) { // prepare to backtrack
            path.pop_back();
            g.vertices[uKey].color = "white";
        }
        return done;
    }
    return true;
}

```

```

bool knightTour(int n, vector<string>& path, string uKey, int limit, Graph&
g) {
    g.vertices[uKey].color = "gray";
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            }
        }
        if (!done) { // prepare to backtrack
            path.pop_back();
            g.vertices[uKey].color = "white";
        }
        return done;
    }
    return true;
}

```

The function is recursive. If the path has reached the required depth, it returns `true`. Otherwise it examines an unvisited neighbor, recurses one level deeper, and returns whether that choice eventually completes the tour.

DFS also uses colors to keep track of which vertices in the graph have been visited. Unvisited vertices are colored white, and visited vertices are colored gray. If all neighbors of a particular vertex have been explored and we have not yet reached our goal length of 64 vertices, we have reached a dead end and must backtrack.

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Backtracking happens when a recursive call returns `false`: the last key is removed from `path` and that vertex is painted white again. The recursive call stack records the choices that remain to be reconsidered.

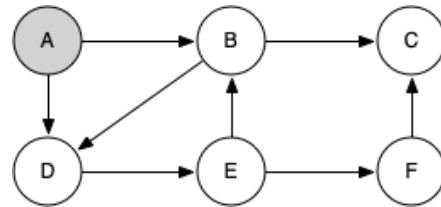
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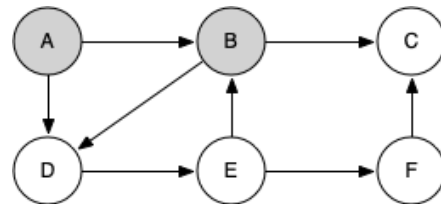
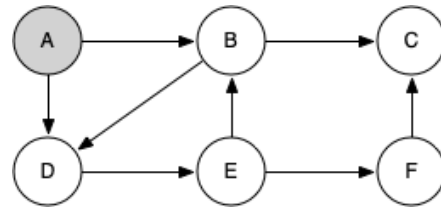
When a recursive call returns `false`, the loop proceeds to the next white neighbor. If none completes the tour, the current call removes its vertex from `path`, resets it to white, and returns `false`.

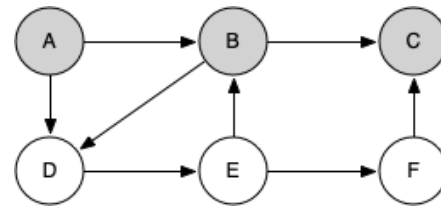
For a small trace, assume iteration over the `neighbors` map follows key order. We call `knightTour(1, path, "A", 6, g)` and watch the recursive choices and backtracking.

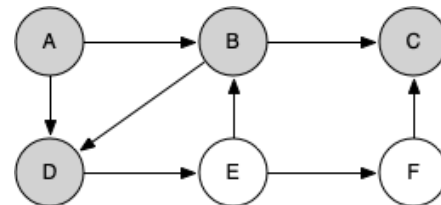
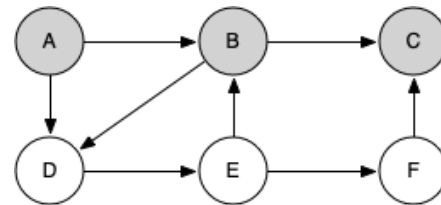
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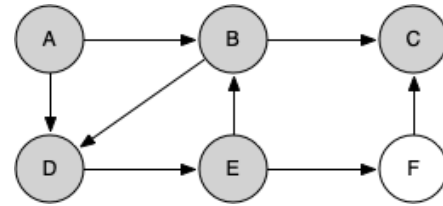


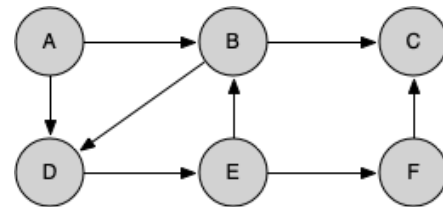
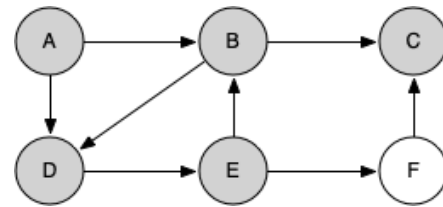
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When the search succeeds, the `vector<string> path` contains `[A, B, D, E, F, C]`, the order that visits each vertex exactly once.

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```
In [5]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // genLegalMoves + knightGraph + ...
using namespace std;

int main() {
    // brute force: 5x5 board - 8x8 without the heuristic takes far too long!
    int boardSize = 5;
    Graph ktGraph = knightGraph(boardSize);
    vector<string> path;
    bool finished = knightTour(1, path, "0", boardSize * boardSize, ktGraph);
    if (finished) printBoard(path, boardSize); // squares show the move number
    else cout << "No path found." << endl;
    return 0;
}
```

0 5 14 11 20

15 10 19 6 13

4 1 12 21 18

9 16 23 2 7

24 3 8 17 22

Instead of a matplotlib figure, we print the board as a grid where each square shows the move number on which the knight visits it — `0` is the start and 24 (or 63) the final move.

For an 8×8 board this plain depth-first search is hopeless — the next section explains why, and the *Warnsdorff heuristic* below fixes it.

Each vertex of the knight graph also carries the usual BFS/DFS bookkeeping state (color, previous), which the tour uses to avoid revisiting squares.

9.14. Knight's Tour Analysis

`knight_tour()` is very sensitive to the method you use to select the next vertex to visit. For example, on a 5×5 board you can produce a path in about 1.5 seconds on a reasonably fast computer. But what happens if you try an 8×8 board? In this case, depending on the speed of your computer, you may have to wait up to several minutes to get the results!

`knight_tour()` is very sensitive to the method you use to select the next vertex to visit. For example, on a 5×5 board you can produce a path in about 1.5 seconds on a reasonably fast computer. But what happens if you try an 8×8 board? In this case, depending on the speed of your computer, you may have to wait up to several minutes to get the results!

The reason for this is that the knight's tour problem as we have implemented it so far is an exponential algorithm of size $O(k^N)$, where N is the number of squares on the chess board, and k is a small constant.

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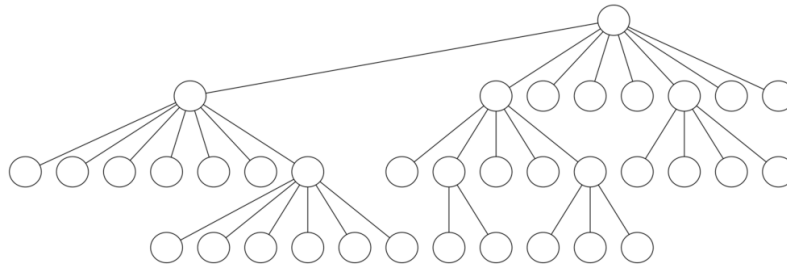
We can use a tree to help us understand this point. The root of the tree represents the starting point of the search. From there the algorithm generates and checks each of the possible moves the knight can make.

As we have noted before, the number of moves possible depends on the position of the knight on the board. In the corners there are only two legal moves, on the squares adjacent to the corners there are three, and in the middle of the board there are eight.

2	3	4	4	4	4	3	2
3	4	6	6	6	6	4	3
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
3	4	6	6	6	6	4	3
2	3	4	4	4	4	3	2

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3	4	6	6	6	6	4	3
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
3	4	6	6	6	6	4	3
2	3	4	4	4	4	3	2

At the next level of the tree there are once again between two and eight possible next moves from the position we are currently exploring. The number of possible positions to examine corresponds to the number of nodes in the search tree.



For a tree with nodes that may have up to eight children, the number of nodes is large. Because the branching factor of each node is variable, we could estimate the number of nodes using an **average branching factor**. The important thing to note is that this algorithm is exponential: $\frac{k^{N+1}-1}{k-1}$, where k is the average branching factor for the board.

Let's look at how rapidly this grows! For a board that is 5×5 the tree will be 25 levels deep. The average branching factor is $k = 3.8$ so the number of nodes in the search tree is $3.8^{25} - 1$ or 3.12×10^{14} .

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For a 6×6 board, $k = 4.4$, there are 1.5×10^{23} nodes, and for a regular 8×8 chess board, $k = 5.25$, there are 1.2×10^{46} !

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For a 6×6 board, $k = 4.4$, there are 1.5×10^{23} nodes, and for a regular 8×8 chess board, $k = 5.25$, there are 1.2×10^{46} !

Of course, since there are multiple solutions to the problem we won't have to explore every single node, but the fractional part of the nodes we do have to explore is just a constant multiplier which does not change the exponential nature of the problem.

`orderByAvail()` orders candidate neighbor keys by their number of remaining white neighbors. Trying the least constrained count first is Warnsdorff's rule and makes the 8×8 case practical, without changing the exponential worst-case guarantee.

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```
vector<string> orderByAvail(Graph& g, string uKey) {
    vector<pair<int, string>> resList;
    for (auto& nb : g.vertices[uKey].neighbors) {
        if (g.vertices[nb.first].color == "white") {
            int c = 0;
            for (auto& w : g.vertices[nb.first].neighbors) {
                if (g.vertices[w.first].color == "white") {
                    c++;
                }
            }
            resList.push_back({c, nb.first});
        }
    }
    sort(resList.begin(), resList.end()); // fewest onward moves first
    vector<string> result;
    for (auto& p : resList) {
        result.push_back(p.second);
    }
    return result;
}
```

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                if (g.vertices[w.first].color == "white") {
                    c++;
                }
            }
            resList.push_back({c, nb.first});
        }
    }
    sort(resList.begin(), resList.end()); // fewest onward moves first
    vector<string> result;
    for (auto& p : resList) {
        result.push_back(p.second);
    }
    return result;
}
```

You might think this is really counterproductive; why not select the node that has the most available moves?

The problem with using the vertex with the most available moves as your next vertex on the path is that it tends to have the knight visit the middle squares early on in the tour!

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When this happens it is easy for the knight to get stranded on one side of the board where it cannot reach unvisited squares on the other side of the board. On the other hand, visiting the squares with the fewest available moves first pushes the knight to visit the squares around the edges of the board first.

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This ensures that the knight will visit the hard-to-reach corners early and can use the middle squares to **hop across the board only when necessary**. Utilizing this kind of knowledge to speed up an algorithm is called a heuristic. This particular heuristic is called Warnsdorff's algorithm.


```

bool knightTour(int n, vector<string>& path, string uKey, int limit, Graph&
g) {
    g.vertices[uKey].color = "gray";
    path.push_back(uKey);
    if (n < limit) {
        bool done = false;
        for (string nbKey : orderByAvail(g, uKey)) {    // Warnsdorff order
            if (done) break;
            if (g.vertices[nbKey].color == "white") {
                done = knightTour(n + 1, path, nbKey, limit, g);
            }
        }
        if (!done) {
            path.pop_back();
            g.vertices[uKey].color = "white";
        }
        return done;
    }
    return true;
}

```

```
In [6]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // orderByAvail + knightTourWarns
using namespace std;

int main() {
    int boardSize = 8; // with the heuristic, 8x8 finishes instantly
    Graph ktGraph = knightGraph(boardSize);
    vector<string> path;
    bool finished = knightTourWarnsdorff(1, path, "0", boardSize * boardSize,
    if (finished) printBoard(path, boardSize);
    else cout << "No path found." << endl;
    return 0;
}
```

0 3 40 19 38 5 42 21

33 18 1 4 41 20 37 6

2 51 34 39 36 55 22 43

17 32 49 62 53 44 7 56

50 13 52 35 48 57 54 23

31 16 61 58 63 26 45 8

12 59 14 29 10 47 24 27

15 30 11 60 25 28 9 46

With Warnsdorff's heuristic the 8×8 tour completes in milliseconds — a dramatic contrast with the exponential brute-force search.

9.15. General Depth-First Search

The knight's tour uses depth-first backtracking to search for one simple path that visits every square. General DFS has a different goal: it explores every reachable vertex and records a depth-first tree; running it from each remaining white vertex produces a depth-first forest, whose trees may have many branches.

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The more general depth-first search is actually easier. Its goal is to search as deeply as possible, connecting as many nodes in the graph as possible and branching where necessary.

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The more general depth-first search is actually easier. Its goal is to search as deeply as possible, connecting as many nodes in the graph as possible and branching where necessary.

It is even possible that a depth-first search will create more than one tree. When the depth-first search algorithm creates a group of trees we call this a **depth-first forest**.

As with the breadth-first search, our depth-first search makes use of predecessor links to construct the tree. In addition, the depth-first search will make use of two additional instance variables in the `Vertex` class. The new instance variables are the discovery and closing times.

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- The closing time is the number of steps in the algorithm before a vertex is colored black.

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- The discovery time tracks the number of steps in the algorithm before a vertex is first encountered.
- The closing time is the number of steps in the algorithm before a vertex is colored black.

As we will see after looking at the algorithm, the discovery and closing times of the nodes provide some interesting properties we can use in later algorithms.

```

class DFSGraph : public Graph {
public:
    int time = 0;
    map<string, int> discovery;
    map<string, int> closing;

    void dfs() {
        for (auto& p : vertices) {
            p.second.color = "white";
            p.second.previous = "";
        }
        for (auto& p : vertices) {
            if (p.second.color == "white") {
                dfsVisit(p.first);
            }
        }
    }
    ...
}

```



```

class DFSGraph : public Graph {
public:
    int time = 0;
    map<string, int> discovery;
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    void dfs() {
        for (auto& p : vertices) {
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        }
        for (auto& p : vertices) {
            if (p.second.color == "white") {
                dfsVisit(p.first);
            }
        }
    }
    ...

    ...
    void dfsVisit(string startKey) {
        vertices[startKey].color = "gray";
        time = time + 1;
        discovery[startKey] = time;
        for (auto& n : vertices[startKey].neighbors) {
            if (vertices[n.first].color == "white") {
                vertices[n.first].previous = startKey;
                dfsVisit(n.first);
            }
        }
    }
}

```

```
vertices[startKey].color = "black";  
time = time + 1;  
closing[startKey] = time;  
}  
};
```

`dfs()` first resets traversal state, then iterates over all vertices and calls `dfsVisit()` on each still-white root. This produces a depth-first forest and also makes repeated runs deterministic.

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The reason we iterate over all the nodes, rather than simply searching from a chosen starting node, is to make sure that all nodes in the graph are considered and that no vertices are left out of the depth-first forest.

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It may look unusual to see the statement `for vertex in self`, but remember that in this case `self` is an instance of the `DFSGraph` class, and iterating over all the vertices in an instance of a graph is a natural thing to do.

`dfsVisit(startKey)` paints one vertex gray and recursively explores its white neighbors as deeply as possible before painting it black.

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`dfsVisit()` and `bfs()` share the same color test. The decisive difference is scheduling: DFS immediately recurses on a white neighbor, while BFS enqueues it for later processing.

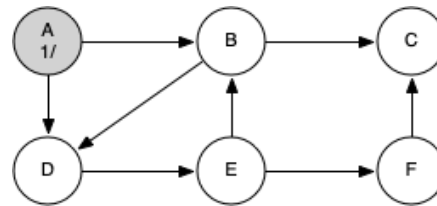
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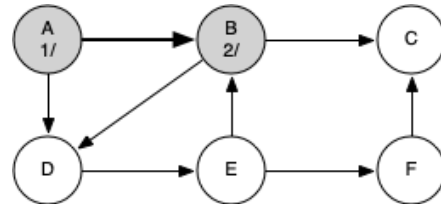
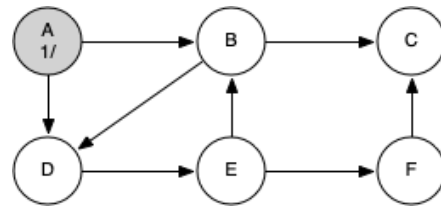
BFS stores pending work in an explicit queue. Recursive `dfsVisit()` uses the C++ call stack implicitly.

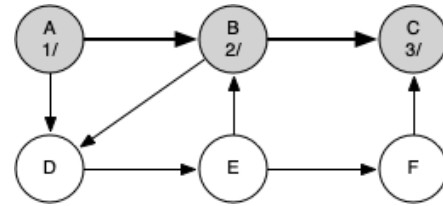
The following sequence of figures illustrates the depth-first search algorithm in action for a small graph. In these figures, the dotted lines indicate edges that are checked, but the node at the other end of the edge has already been added to the depth-first tree.

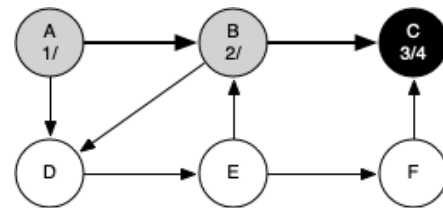
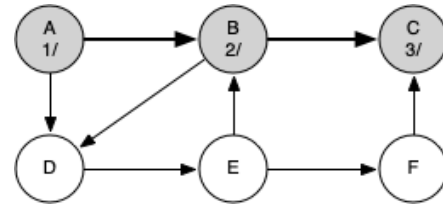
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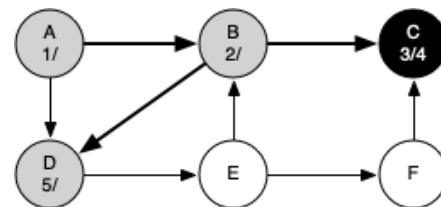
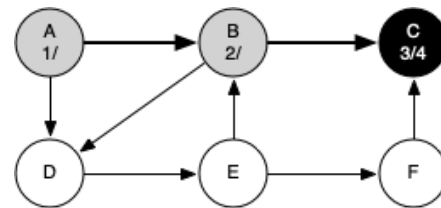
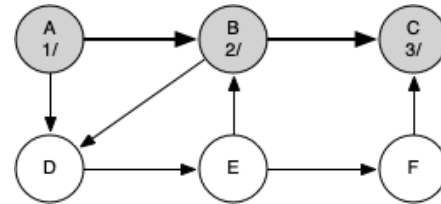


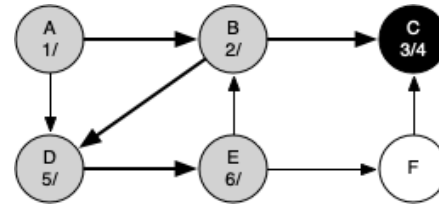
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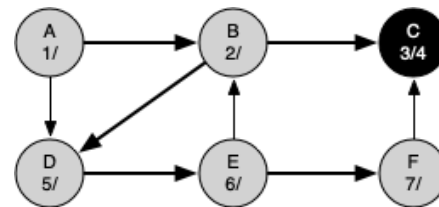
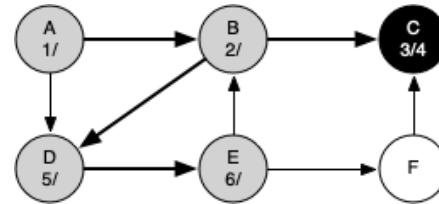


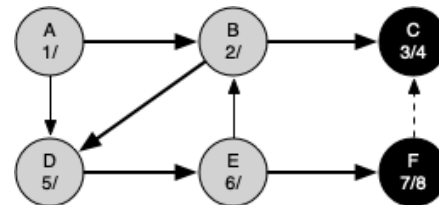
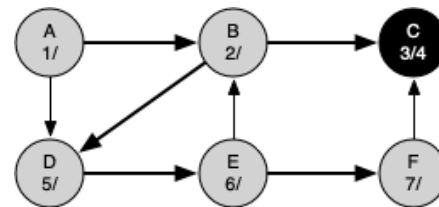
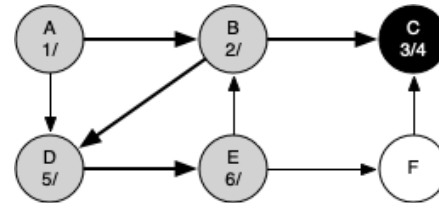


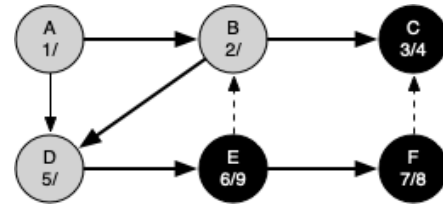


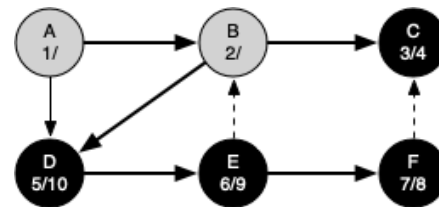
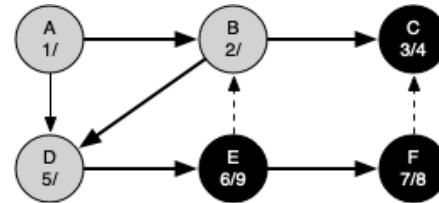


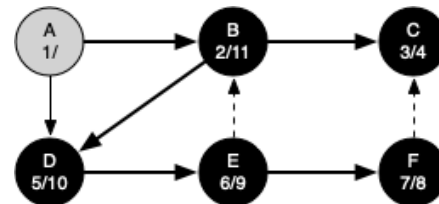
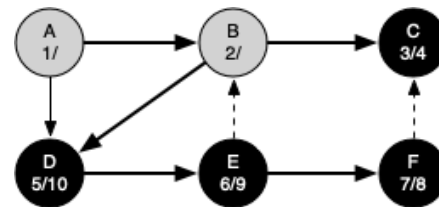
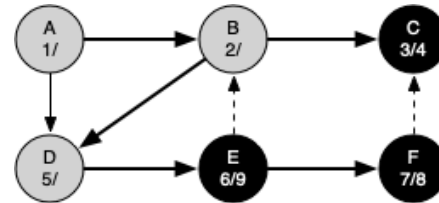


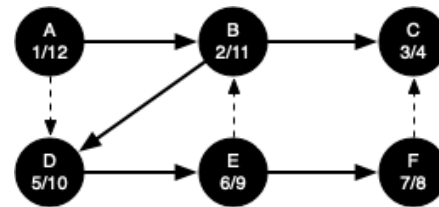


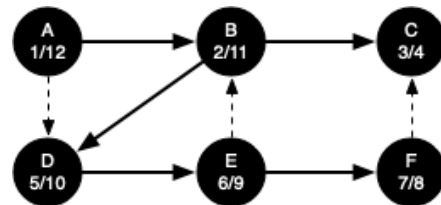












The discovery and closing times for each node display a property called the parenthesis property. This property means that all the children of a particular node in the depth-first tree have a later discovery time and an earlier closing time than their parent.

In [7]:

```
#include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // DFSGraph (md listing above)
using namespace std;

int main() {
    DFSGraph g;
    g.addEdge("A", "B"); g.addEdge("B", "C");
    g.addEdge("A", "D"); g.addEdge("B", "D");
    g.addEdge("D", "E"); g.addEdge("E", "B");
    g.addEdge("E", "F"); g.addEdge("F", "C");
    g.dfs();
    printf("%4s|%9s|%8s|%9s\n", "Key", "Discover", "Closing", "Previous");
    for (auto& p : g.vertices)
        printf("%4s|%9d|%8d|%9s\n", p.first.c_str(),
            g.discovery[p.first], g.closing[p.first], p.second.previous.c_s
    return 0;
}
```

Key| Discover| Closing| Previous

A| 1| 12|

B| 2| 11| A

C| 3| 4| B

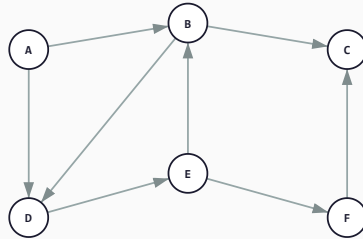
D| 5| 10| B

E| 6| 9| D

F| 7| 8| E

```
In [ ]: load_anim("dfs")
```

DFS 深度優先搜尋 — 一條路走到黑，碰壁再回頭



› 按「開始」跑完整 DFS (會走遍整個 forest)

起點

A ▼

速度



▶ 開始

→ 單步

↺ 重置

Recursion stack (隱式): empty

時間計數: 0

頂點時間表

v	color	disc	close	prev
A	white	—	—	—
B	white	—	—	—
C	white	—	—	—
D	white	—	—	—
E	white	—	—	—
F	white	—	—	—

9.16. Depth-First Search Analysis

The outer loops in `dfs()` take $O(|V|)$ excluding recursive edge scans: reset visits every vertex once, and the forest loop considers every vertex as a possible root once.

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In `dfsVisit()`, adjacency scans total $O(|E|)$ because each directed adjacency entry is examined once from its source. Therefore DFS takes $O(|V| + |E|)$ time.


```
In [ ]: display_quiz(quiz_path+"bfs_dfs.json", max_width=800)
```

What is the primary structural difference in how Breadth-First Search (BFS) and Depth-First Search (DFS) track the next vertex to explore?

- ☐ BFS uses a queue to explore layer by layer, while DFS uses a stack (often via recursion) to explore as deep as possible.
- ☐ BFS uses a stack while DFS uses a queue.
- ☐ BFS tracks vertices using a map, while DFS only uses a simple vector.
- ☐ Both algorithms must use a priority queue to ensure the correct search order.

9.17. Topological Sorting

Course status: optional. Topological sorting is an official cppds section; this concise summary remains visible in RISE while the detailed derivation stays optional.

To demonstrate that computer scientists can turn just about anything into a graph problem, let's consider the difficult problem of stirring up a batch of pancakes.

When the pancakes start to bubble you turn them over and let them cook until they are golden brown on the bottom. Before you eat your pancakes you are going to want to heat up some syrup.

9.18. Strongly Connected Components

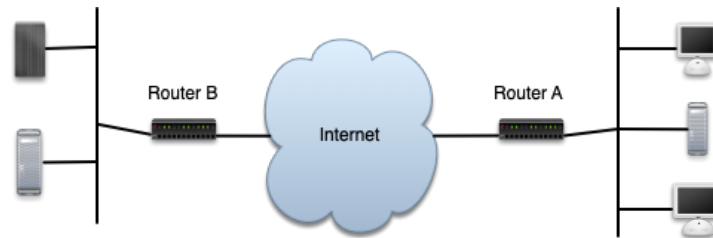
Course status: optional. Strongly connected components are an official cppds section; this concise summary remains visible in RISE while the detailed derivation stays optional.

For the remainder of this chapter we turn to graphs formed by links between web pages and hosts on the internet.

9.19. Shortest Path Problems

When you surf the web, send an email, or log in to a laboratory computer from another location on campus, a lot of work is going on behind the scenes to get the information on your computer transferred to another computer. The in-depth study of how information flows from one computer to another over the internet is the primary topic for a class in computer networking.

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When you surf the web, send an email, or log in to a laboratory computer from another location on campus, a lot of work is going on behind the scenes to get the information on your computer transferred to another computer. The in-depth study of how information flows from one computer to another over the internet is the primary topic for a class in computer networking.

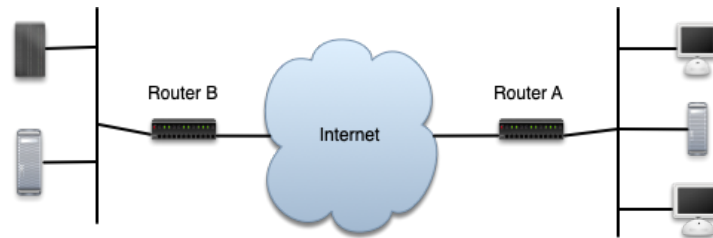


Figure above shows you a high-level overview of how communication on the internet works. When you use your browser to request a web page from a server, the request must travel over your local area network and out onto the internet through a router.

Each router on the internet is connected to one or more other routers. If you run the `tracert` command at different times of the day, you are likely to see that your information flows through different routers at different times. This is because there is a cost associated with each connection between a pair of routers that depends on the volume of traffic, the time of day, and many other factors.

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By this time it will not surprise you to learn that we can represent the network of routers as a graph with weighted edges.

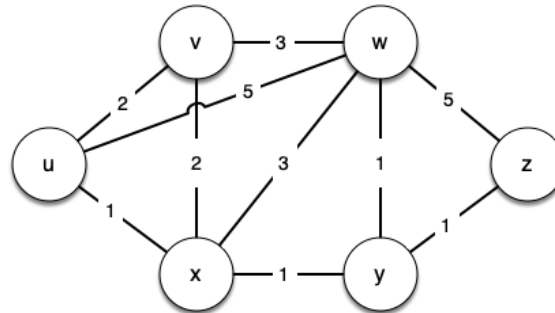
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By this time it will not surprise you to learn that we can represent the network of routers as a graph with weighted edges.

Figure below shows a small example of a weighted graph that represents the interconnection of routers in the internet. The problem that we want to solve is to find the shortest path, one with the smallest total weight along which to route any given message.

This problem should sound familiar because it is similar to the problem we solved using a breadth-first search, except that here we are concerned with the total weight of the path rather than the number of hops in the path. It should be noted that if all the weights are equal, the problem is the same!

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```
In [ ]: display_quiz(quiz_path+"bfs.json", max_width=800)
```

Why is Breadth-First Search (BFS) guaranteed to find the shortest path in an unweighted graph?

- ☐ It always chooses the edge with the smallest numerical weight first.
- ☐ It explores all vertices at distance k from the starting vertex before moving to vertices at distance $k+1$.
- ☐ It marks vertices as black only when the shortest path has been confirmed.
- ☐ It uses a greedy approach to prune paths that are too long.

9.20. Dijkstra's Algorithm

Dijkstra's algorithm is an iterative algorithm that provides us with the shortest path from one particular starting node to all other nodes in the graph. Again this is similar to the results of a breadth-first search.

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The algorithm iterates once for every vertex in the graph; however, the order that it iterates over the vertices is controlled by a priority queue. The value that is used to determine the order of the objects in the priority queue is **distance**.

A priority queue acts like a queue in that you dequeue an item by removing it from the front. However, in a priority queue the logical order of items inside a queue is determined by their priority. The highest priority items are at the front of the queue and the lowest priority items are at the back.

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Thus when you enqueue an item on a priority queue, the new item may move all the way to the front.

When a vertex is first created, distance is set to a very large number. Theoretically you would set distance to infinity, but in practice we just set it to a number that is larger than any real distance we would have in the problem we are trying to solve.

Our C++ implementation uses a **lazy priority queue**. When relaxation improves a distance, it pushes a new `(distance, key)` pair instead of searching the heap for the old entry. When an older, larger entry reaches the top, the algorithm recognizes it as stale and skips it.

The priority queue can therefore contain more than one entry for a vertex. Every entry carries both the tentative distance and the vertex key, so popping the minimum entry still identifies which graph vertex to process.

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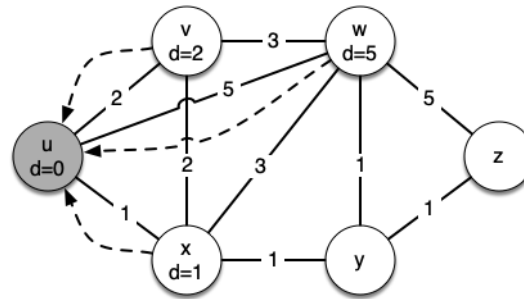
There is no `changePriority()` call in this version. Each successful relaxation performs one heap push, so there are at most $O(|E|)$ pushed entries and the same number of pops.

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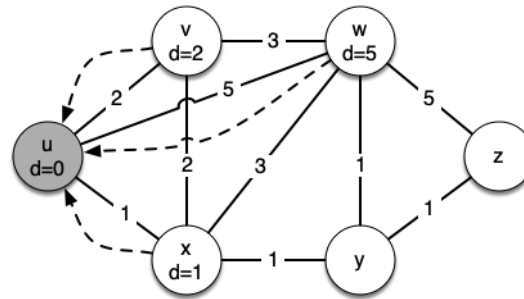
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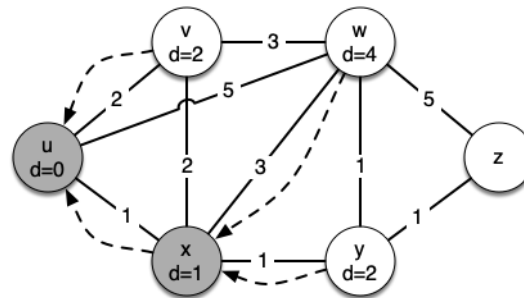
Start at u with `distance = 0`; every other vertex begins at `INT_MAX` (unreachable). Relaxing u 's incident edges assigns their first finite candidate distances and records u as predecessor.



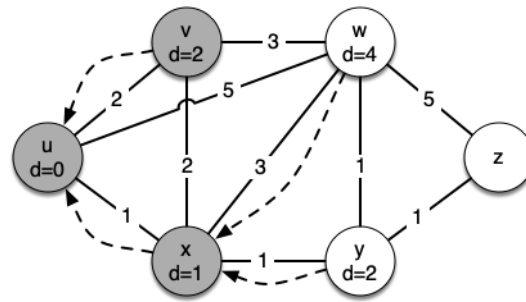
PQ = x, v, w



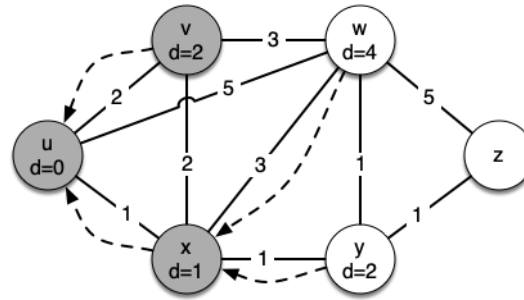
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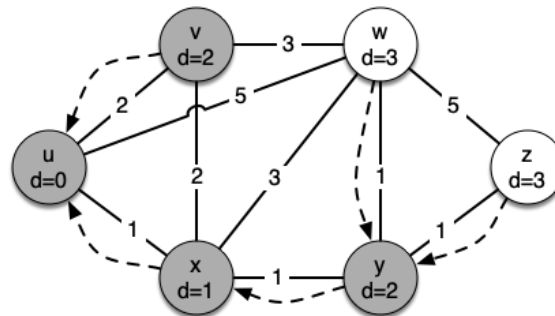
PQ = v, y, w



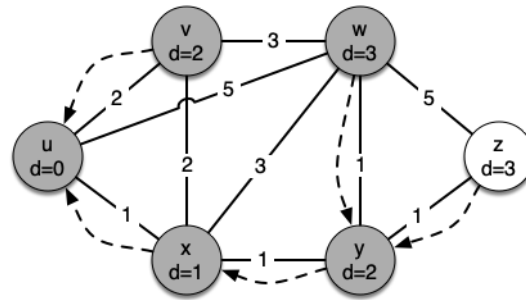
PQ = y, w



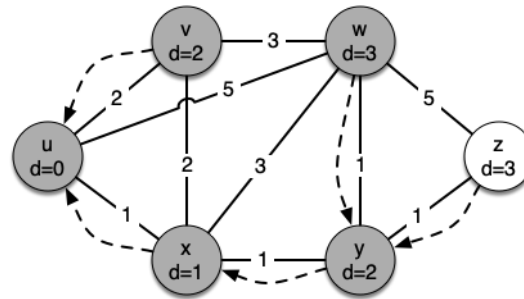
PQ = y, w



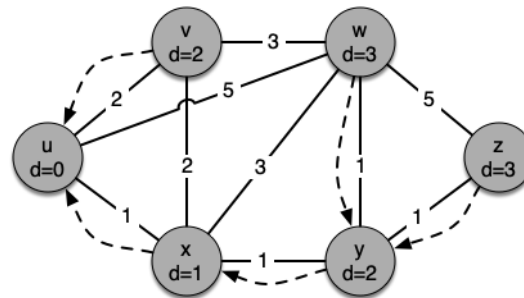
PQ = w, z



PQ = z



PQ = z



PQ = None

```

In [8]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // dijkstra (md listing above)
using namespace std;

int main() {
    Graph g;
    g.addEdge("u","v",2); g.addEdge("v","u",2); g.addEdge("v","w",3); g.addEdge
    g.addEdge("w","z",5); g.addEdge("z","w",5); g.addEdge("u","x",1); g.addEdge
    g.addEdge("u","w",5); g.addEdge("w","u",5); g.addEdge("x","v",2); g.addEdge
    g.addEdge("x","w",3); g.addEdge("w","x",3); g.addEdge("x","y",1); g.addEdge
    g.addEdge("y","w",1); g.addEdge("w","y",1); g.addEdge("y","z",1); g.addEdge
    dijkstra(g, "u");
    for (auto& p : g.vertices) // key: distance (previous)
        cout << p.first << ": " << p.second.distance
            << " (" << p.second.previous << ")" << endl;
    return 0;
}

```

u: 0 ()

v: 2 (u)

w: 3 (y)

x: 1 (u)

y: 2 (x)

z: 3 (y)

Every vertex now carries the length of the **shortest** path from **u**, and following **previous** pointers backwards reconstructs that path.

```

In [9]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp"    // dijkstra + findPath
using namespace std;

int main() {
    Graph g;
    g.addEdge("u", "v", 2); g.addEdge("v", "u", 2); g.addEdge("v", "w", 3); g.addEdge
    g.addEdge("w", "z", 5); g.addEdge("z", "w", 5); g.addEdge("u", "x", 1); g.addEdge
    g.addEdge("u", "w", 5); g.addEdge("w", "u", 5); g.addEdge("x", "v", 2); g.addEdge
    g.addEdge("x", "w", 3); g.addEdge("w", "x", 3); g.addEdge("x", "y", 1); g.addEdge
    g.addEdge("y", "w", 1); g.addEdge("w", "y", 1); g.addEdge("y", "z", 1); g.addEdge
    dijkstra(g, "u");
    for (string v : findPath(g, "u", "z")) cout << v << " ";
    cout << endl;
    return 0;
}

```

u x y z

Dijkstra's correctness requires every edge weight to be **non-negative**; zero-weight edges are valid. A negative edge can reveal a shorter route after a vertex has been treated as final, so the greedy proof no longer applies. It does not imply that the program must loop forever: the result is simply not guaranteed. The course implementation rejects negative weights.

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We will note that to route messages through the internet, other algorithms are used for finding the shortest path. One of the problems with using Dijkstra's algorithm on the internet is that you must have a complete representation of the graph in order for the algorithm to run. The implication of this is that every router has a complete map of all the routers in the internet!

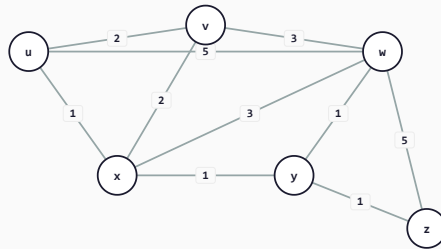
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In practice this is not the case and other variations of the algorithm allow each router to discover the graph as they go.


```
In [ ]: load_anim("dijkstra")
```

Dijkstra 最短路徑 — 加權圖上的「貪婪式 BFS」



› 選擇起點，按「開始」執行 Dijkstra

起點

u ▼

速度



▶ 開始

→ 單步

↺ 重置

Priority Queue :

empty

距離表

LIVE

v	distance	prev	visited
u	∞	—	
v	∞	—	
w	∞	—	
x	∞	—	
y	∞	—	
z	∞	—	

9.21. Analysis of Dijkstra's Algorithm

In the lazy C++ implementation, resetting all vertex distances costs $O(|V|)$ and only $(0, \text{start})$ is initially pushed. Scanning adjacency maps examines each directed edge once during the effective expansions.

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Each successful relaxation adds one heap entry, so there are at most $O(|E|)$ pushes and pops. A heap operation costs $O(\log |E|) = O(\log |V|)$ for this simple map-backed graph, and ordered-map vertex lookup contributes the same logarithmic factor. The total is $O((|V| + |E|) \log |V|)$, with $O(|V| + |E|)$ storage including stale entries.

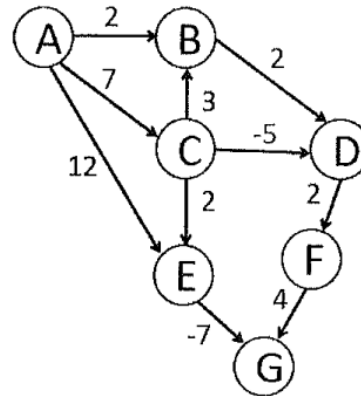
```
In [ ]: display_quiz(quiz_path+"dijk.json", max_width=800)
```

Which of the following is a known limitation or characteristic of Dijkstra's algorithm?

- ☐ It is an iterative algorithm that provides the shortest path from a single source to all other reachable vertices.
- ☐ The time complexity is independent of the number of edges in the graph.
- ☐ It does not require a complete map of the graph to function in a routing environment.
- ☐ It can accurately find shortest paths in graphs with negative edge weights.

Exercise 1: Negative-edge precondition

The pictured graph contains negative-weight edges. What happens if it is passed to the course `dijkstra(g, "A")` implementation?



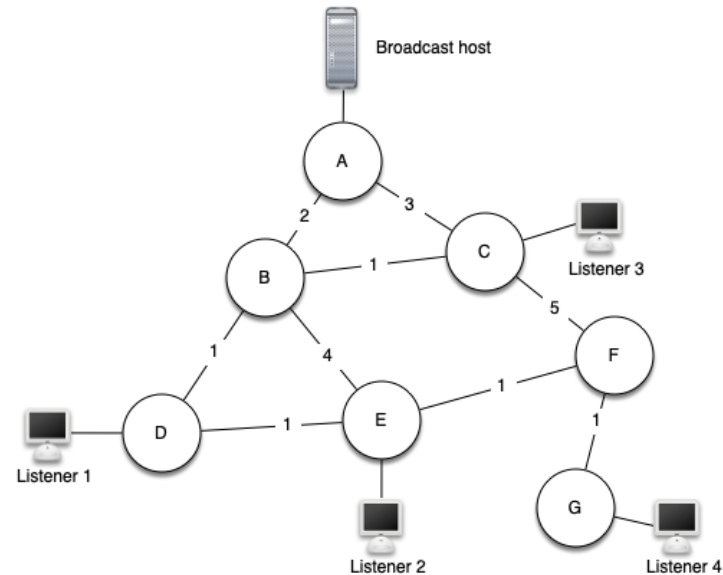
- (A) It returns `A->B->D`.
- (B) It returns `A->C->D`.
- (C) It silently deletes the negative edges.
- (D) It throws `invalid_argument` before running the priority-queue traversal.

Answer: (D). Lazy duplicate entries do not make Dijkstra valid for negative weights; the implementation explicitly enforces the non-negative precondition.

9.22. Prim's Spanning Tree Algorithm

For our last graph algorithm let's consider a problem that online game designers. The problem is that they want to efficiently transfer a piece of information to anyone and everyone who may be listening. This is important in gaming so that all the players know the very latest position of every other player!

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In the above figure we show a small network with a broadcaster and some listeners. Using this first approach, four copies of every message would be sent. Assuming that the **least cost path** is used, let's see how many times each router would handle the same message.

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All messages from the broadcaster go through router A, so A sees all four copies of every message. Router C sees only one copy of each message for its listener. However, routers B and D would see three copies of every message since routers B and D are on the **cheapest path** for listeners 1, 2, and 4! When you consider that the broadcast host must send hundreds of messages each second for a radio broadcast, that is a lot of extra traffic.

Another brute force solution is for the broadcast host to send a single copy of the broadcast message and let the routers sort things out. In this case, the easiest solution is a strategy called uncontrolled flooding:

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Each message starts with a time to live (TTL) value set to some number greater than or equal to the number of edges between the broadcast host and its most distant listener. Each router gets a copy of the message and passes the message on to all of its neighboring routers. When the message is passed on the TTL is decreased. Because each router continues to send copies of the message to all its neighbors until the TTL value reaches 0, it is easy to convince yourself that uncontrolled flooding generates many more unnecessary messages than our first strategy.

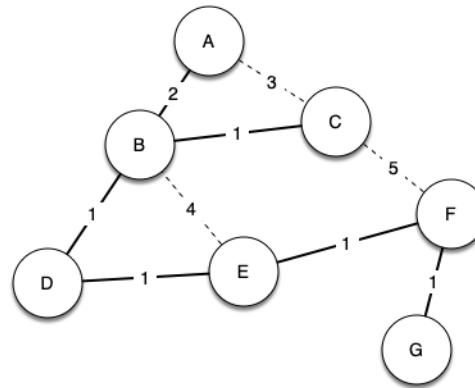
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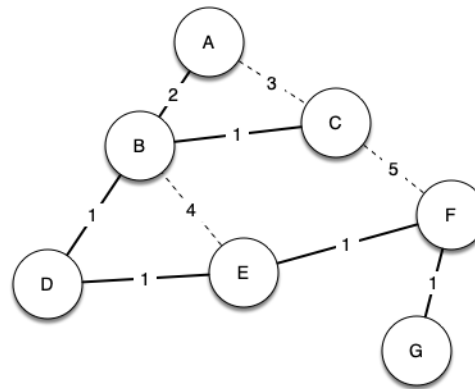
The solution to this problem lies in the construction of a minimum weight spanning tree. Formally we define the minimum spanning tree T for a graph $G = (V, E)$ as follows. T is an acyclic subset of E that connects all the vertices in V . The sum of the weights of the edges in T is minimized.

Below shows a simplified version of the broadcast graph and highlights the edges that form a minimum spanning tree for the graph.

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In this example A forwards the message to B. B forwards the message to D and C. D forwards the message to E, which forwards it to F, which forwards it to G. No router sees more than one copy of any message, and all the listeners that are interested see a copy of the message.

Prim's algorithm is a greedy algorithm for a weighted **undirected** graph. At each step it chooses a **safe edge**: a minimum-weight edge crossing the cut between vertices already in the growing tree and vertices still outside it. The cheapest edge elsewhere in the graph is not necessarily eligible.

The course `prim()` requires one connected component containing every vertex. After the lazy priority queue empties, it compares the number of vertices added to the tree with `g.vertices.size()` and throws `invalid_argument` if the graph is disconnected.

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While T is not yet a spanning tree
  Find an edge that is safe to add to the tree
  Add the new edge to T
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```

The key step is to find a safe edge: among all edges crossing from the current tree to a vertex outside it, Prim's algorithm chooses one with minimum weight. Crossing the cut preserves connectivity without creating a cycle, and the cut property guarantees that this minimum-weight choice can belong to an MST.

```
void prim(Graph& g, string startKey) {
    priority_queue<pair<int, string>,
                  vector<pair<int, string>>,
                  greater<pair<int, string>>> pq;
    set<string> inTree;

    for (auto& p : g.vertices) {
        p.second.distance = INT_MAX;
        p.second.previous = "";
    }
    g.vertices[startKey].distance = 0;
    pq.push({0, startKey});
    ...
}
```



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    }
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    pq.push({0, startKey});
    ...

    ...
    while (!pq.empty()) {
        auto [distance, currentKey] = pq.top();
        pq.pop();
        if (inTree.count(currentKey)) continue;    // stale entry
        inTree.insert(currentKey);
        for (auto& n : g.vertices[currentKey].neighbors) {
            int newDistance = n.second;    // edge weight only, not path
            if (!inTree.count(n.first) && newDistance <
                g.vertices[n.first].distance) {
                g.vertices[n.first].distance = newDistance;
                g.vertices[n.first].previous = currentKey;
                pq.push({newDistance, n.first});
            }
        }
    }
}

```

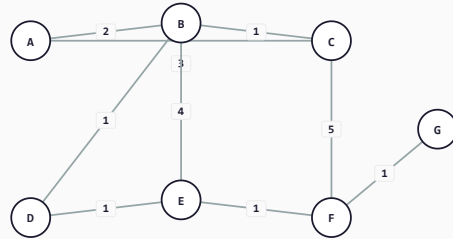


```
}  
if (inTree.size() != g.vertices.size())  
    throw invalid_argument("Prim requires a connected graph");  
}
```

(complete listing: `pythonds3/cppds/graph_algos.hpp`)

```
In [ ]: load_anim("prim")
```

Prim 最小生成樹 — 用最小邊權連通整張圖



› 選擇起點，按「開始」逐步建立 MST

起點

A ▼

速度



▶ 開始

→ 單步

↺ 重置

Priority Queue :

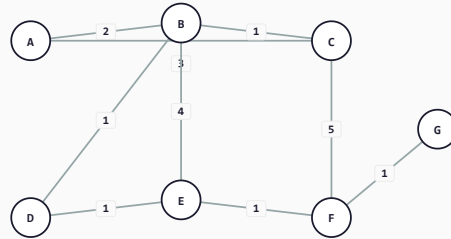
empty

MST 邊清單

—

```
In [ ]: load_anim("prim")
```

Prim 最小生成樹 — 用最小邊權連通整張圖

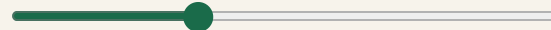


› 選擇起點，按「開始」逐步建立 MST

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A ▼

速度



▶ 開始

→ 單步

↺ 重置

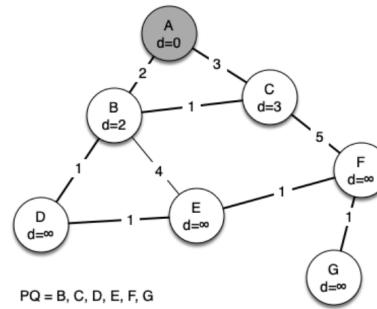
Priority Queue:

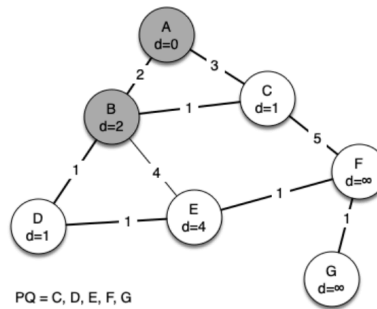
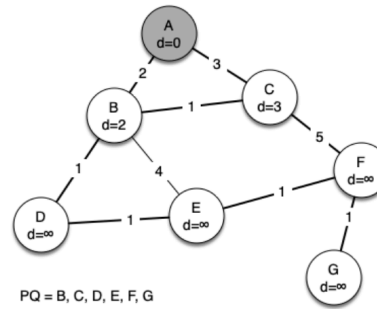
empty

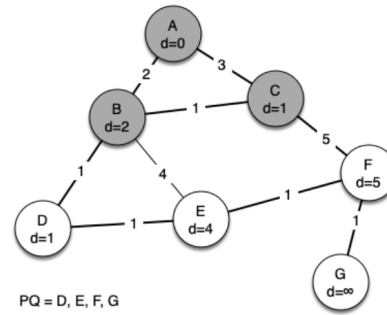
MST 邊清單

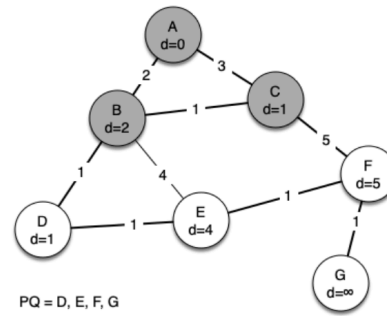
—

The following sequence of figures shows the algorithm in operation on our sample tree. We begin with the starting vertex as A. The distances to all the other vertices are initialized to infinity.

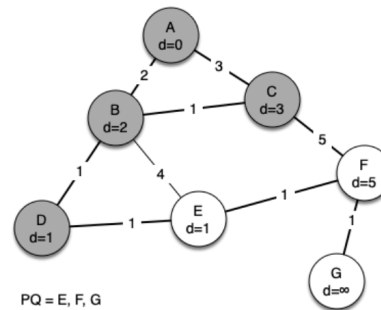




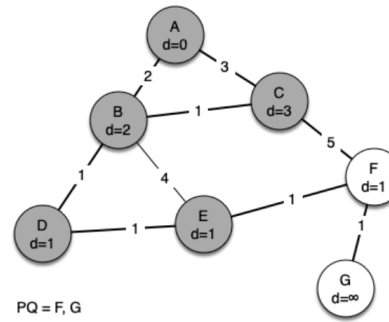


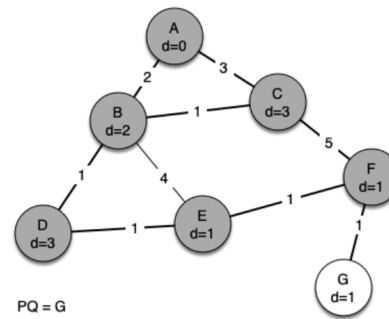
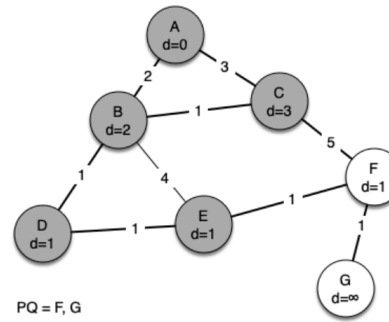


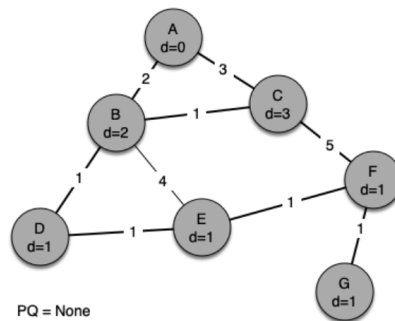
PQ = D, E, F, G



PQ = E, F, G







```

In [10]: #include <iostream>
#include "pythonds3/cppds/graph_algos.hpp" // prim (md listing above)
using namespace std;

int main() {
    Graph g;
    g.addEdge("A", "B", 2); g.addEdge("B", "A", 2); g.addEdge("A", "C", 3); g.addEdge
    g.addEdge("B", "D", 1); g.addEdge("D", "B", 1); g.addEdge("B", "C", 1); g.addEdge
    g.addEdge("B", "E", 4); g.addEdge("E", "B", 4); g.addEdge("D", "E", 1); g.addEdge
    g.addEdge("C", "F", 5); g.addEdge("F", "C", 5); g.addEdge("E", "F", 1); g.addEdge
    g.addEdge("F", "G", 1); g.addEdge("G", "F", 1);

    prim(g, "A");
    cout << "MST edges: ";
    for (auto& p : g.vertices)
        if (p.second.previous != "")
            cout << "(" << p.second.previous << ", " << p.first << ") ";
    cout << endl;
    return 0;
}

```

MST edges: (A,B) (B,C) (B,D) (D,E) (E,F) (F,G)

The printed edges form the minimum spanning tree rooted at **A**, discovered by repeatedly choosing the cheapest edge that crosses the current tree boundary.

```
In [ ]: display_quiz(quiz_path+"prim.json", max_width=800)
```

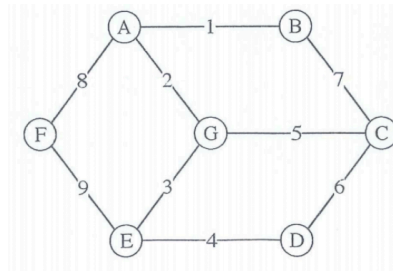
In Prim's algorithm for finding a Minimum Spanning Tree (MST), what defines a safe edge?

- ☐ Any edge that does not create a cycle when added to the current tree.
- ☐ A minimum-weight edge crossing from the current tree to a vertex outside it.
- ☐ The first edge encountered during a depth-first search of the graph.
- ☐ An edge that has a weight lower than the average of all edges in the graph.

What does the course `prim(g, start)` function do when the graph is disconnected?

- ☐ It throws `invalid_argument` after detecting that not every vertex entered the tree.
- ☐ It silently returns a minimum spanning forest.
- ☐ It adds arbitrary edges between disconnected components.
- ☐ It loops forever waiting for another priority-queue entry.

Exercise 2: Considering the following graph, when Prim's algorithm is used and starts from node F, what is the weight of the last edge to be added into the minimum spanning tree?



```
In [ ]: from jupytercards import display_flashcards  
display_flashcards("flashcards/ch8.json")
```

Graph



Next

>

References

1. Textbook: *Problem Solving with Algorithms and Data Structures using C++* (cppds), Chapter 9 (Graphs and Graph Algorithms) —
<https://runestone.academy/ns/books/published/cppds/index.html>

